

Use Case

UC-03: Perform TLS/SSL Audit

Preconditions

1. The SysAdmin has added one or more SSL/TLS endpoints to the monitoring system.
2. The monitored endpoint is active and reachable.
3. The daily monitoring schedule is configured.

Postconditions

1. The SSL/TLS certificate expiry information is recorded.
2. An expiry alert is generated if the certificate reaches an alert threshold.
3. The audit result is stored in the system log.

Main Success Scenario

1. The scheduled daily monitoring process starts.
2. The system selects a monitored SSL/TLS endpoint.
3. The system initiates a TLS handshake with the endpoint.
4. The system retrieves the SSL/TLS certificate.
5. The system extracts the certificate expiration date.
6. The system calculates the number of days remaining before expiry.
7. The system checks whether the remaining days match an alert threshold of **30, 15, or 3 days**.
8. If an alert threshold is reached, the system generates an expiry alert.
9. The system sends the alert to the SysAdmin.
10. The system records the audit result.
11. The system continues with the next monitored endpoint.
12. The use case ends successfully.

Alternate Flow: TLS Handshake Fails

3a. TLS handshake cannot be completed.

1. The system records the handshake failure in the audit log.

2. The system marks the endpoint audit as failed.
3. The system continues monitoring the remaining endpoints.
4. The use case ends for the failed endpoint.